QUIZ 08, VERSION A – Solutions

STAT 1110: Section – Fall 2026 with Dr. Ethan P. Marzban

NAME: _____

First and Last

USER ID: _____

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Instructions:

- You will have **30 minutes** to complete this quiz that is worth **20 points**.
  - Please make note of the point values for each question
- The use of calculators is permitted, but the use of any other aids (including, but not limited to: laptops, cell phones, and note sheets) is prohibited.
- This quiz contains a mix of free-response and multiple-choice questions.
  - Answers to Free-Response Questions should go on the **Answer Page**
  - Answers to Multiple-Choice Questions should go on the **Bubble Sheet**
  - You may write on this exam, but **I will not grade anything written here**.
- **Show your work** on all free-response questions; correct answers with no justification will not receive full marks, whereas partial credit is available for incorrect answers with correct justification.
  - Partial credit is **not** available for multiple-choice questions.
- Ensure your **Name**, **User ID**, and **Version** are marked on the bubble sheet, answer page, and exam booklet.
  - Failure to include this information on any of the three exam components will result in your **quiz not being graded**.
- Good Luck!

Honor Statement

In signing my name below, I certify that all work appearing on this quiz is entirely my own and not copied from any external source. I also certify that I have not received any unauthorized aid (including generative AI tools), and that I have acted in accordance with the Cal Poly standards of academic integrity. I acknowledge that failure to abide by these terms will be treated as academic misconduct.

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Scenario 1: For Multiple Choice Questions 1 - 5 and Free-Response Questions 1 - 2.

Is the prevalence of bike ownership different across different class standings? To answer this question, Ayoade took a simple random sample of 50 Cal Poly Students and recorded their **standing** (Freshman, Sophomore, Junior, or Senior) and their **bike ownership** (whether or not they own a bike; treat "owning a bike" as a "success"). Shown below are some summaries of Ayoade's data.

	Freshman	Sophomore	Junior	Senior	Total
Num. Bike Owners	5	10	6	5	26
Total	10	20	10	10	50

Table 1: Summary of Ayoade's data

MULTIPLE CHOICE QUESTION 1 (1 point)

What is one observational unit in the context of this study?

(**A**) One student who owns a bike

(**B**) One student who does not own a bike

(C) One student

(**D**) Cal Poly Students

(**E**) None of the above

MULTIPLE CHOICE QUESTION 2 (1 point)

When writing down the null hypothesis in symbols and math only, how many parameters do we need?

(**A**) 1 (**B**) 2 (**C**) 3 **(D) 4** (**E**) 5

Solution: We need four parameters (in particular, four population proportions): one for each of the four class standings.

MULTIPLE CHOICE QUESTION 3 (1 point)

True or False: it is possible to write the alternative hypothesis in symbols and math only.

(**A**) True **(B) False**

Solution: Our alternative can only be written verbally, as "at least one class standing has a different population proportion of bike owners than the others."

FREE-RESPONSE QUESTION 1 (2 points)

For your convenience, I've gone ahead and calculated several of the differences needed to calculate the mean group difference statistic.

Groups Diff.	Difference in Sample Proportions
Freshman – Sophomore	0.00
Freshman – Junior	–0.10
Freshman – Senior	0.00
Sophomore – Junior	–0.10
Sophomore – Senior	0.00
Junior – Senior	0.10

Table 2: Partial Calculations

Calculate the value of the mean group difference statistic.

Solution:

$$\text{Mean Group Diff} = \frac{1}{6}\left(|0.00| + |-0.10| + |0.00| + |-0.10| + |0.00| + |0.10|\right) = \frac{1}{6}(0.3) = \mathbf{0.05}$$

MULTIPLE CHOICE QUESTION 4 (1 point)

Suppose Ayoade has calculated the value of the mean group difference statistic to be 0.100 (which is not necessarily the correct answer to Free-Response Question 1 above). Which of the following correctly describes how Ayoade can obtain a simulation-based *p*-value?

(A) Go to the Multiple Proportions applet, input the data, and simulate at least 10,000 shuffles. Then, calculate the proportion of repetitions that gave rise to a Mean Group Difference statistic value of 0.100 or greater.

(**B**) Go to the Multiple Proportions applet, input the data, and simulate at least 10,000 shuffles. Then, calculate the proportion of repetitions that gave rise to a Mean Group Difference statistic value of strictly greater than 0.100.

(**C**) Go to the Multiple Proportions applet, input the data, and simulate at least 10,000 shuffles. Then, calculate the proportion of repetitions that gave rise to a Mean Group Difference statistic value of 0.100 or less.

(**D**) Go to the Multiple Proportions applet, input the data, and simulate at least 10,000 shuffles. Then, calculate the proportion of repetitions that gave rise to a Mean Group Difference statistic value of strictly less than 0.100.

(**E**) Go to the Multiple Proportions applet, input the data, and simulate at least 10,000 shuffles. Then, calculate the proportion of repetitions that gave rise to a Mean Group Difference statistic value of 0.100 or more extreme, in both directions.

Solution: This was designed to test a couple of things:

- Remember that when using the Mean Group Diff statistic we calculate the *p*-value using right-tails
- Remember the "as" part of the "as or more extreme" part of the definition of the *p*-value.

MULTIPLE CHOICE QUESTION 5 (1 point)

Ayoade has taken another (completely different) sample, and obtained a p-value of 0.000. Which of the following is the most correct conclusion that can be drawn, using this new dataset and p-value?

(**A**) There is evidence to suggest that no two class standings have the same population proportions of bike ownership

(B) There is evidence to suggest that at least one class standing has a different population proportion of bike ownership than the other class standings.

(**C**) There is evidence to suggest that all four class standings have the same population proportion of bike ownership

Solution: This was designed to test a couple of things:

- A very small p-value leads us to reject the null hypothesis in favor of the alternative hypothesis
- The alternative hypothesis in a study like this is "there is an association," which means "at least one of the population/long-run proportions is different from the others."

FREE-RESPONSE QUESTION 2 (2 points)

Explain why Ayoade should NOT have conducted pairwise tests of the population proportions of bike ownerships across the different standings. Explain why Ayoade should NOT have conducted pairwise tests of the population proportions of bike ownerships across the different standings. **Note:** there is a **specific term** I'm looking for here - make sure your answer includes this term, defines it, and relates it to the question at hand.

Solution: Multiple Testing tells us that the more tests we conduct on the same dataset, the higher our chances become of committing at least one Type I error. As such, conducting unnecessary pairwise tests unnecessarily increases our chances of committing at least one Type I error; it's better to do one single simultaneous test.

Scenario 2: For Multiple Choice Questions 6 - 7 and Free-Response Questions 3 - 4.

They say that music can make or break a party. But is that true? In particular, **is there an association between the type of music played at a party and partygoers' overall enjoyment of the party?** Anders, a party planner, wants to address this statistically. He recruited 30 volunteers (aged 21 - 34 years), and had each volunteer attend two parties. One party had techno music playing; the other party had pop music playing. At the end of each party, Anders asked every volunteer to rate their overall experience at the party on a scale of 1 to 10, 1 being "I hated it" and 10 being "I loved it."

We'll call the two variables "**music**" (either "techno" or "pop") and "**enjoyment**" (the 10-point scale enjoyment rating). Whenever necessary, we'll use the order of subtraction (Techno — Pop).

MULTIPLE CHOICE QUESTION 6 (1 point)

Which of the following best explains why this data should be analyzed as paired data?

(**A**) Because there are two groups we are comparing across

(**B**) Because both techno and pop are genres of music, so there is a potential for confounding

(**C**) Because the response values (enjoyment) are quantitative

(D) Because the response values (enjoyment) from the two groups come from the same person

FREE-RESPONSE QUESTION 3 (4 points)

Write down the null and alternative hypotheses in symbols and math only, but be sure to define any symbols you use.

Solution: Define μ_d to be the population average difference (Techno – Pop) in enjoyment ratings of a single partygoer. Then our null and alternative hypotheses can be written as

$$\begin{cases} H_0: & \mu_d = 0 \\ H_A: & \mu_d \neq 0 \end{cases}$$

Remember that because the data is paired, we do not define two separate population means; rather, we simply define one population mean *difference*.

MULTIPLE CHOICE QUESTION 7 (1 point)

Which of the following is the correct formula to use when using a theory-based approach to calculate the standardized statistic?

(**A**) $z = \dfrac{(\hat{p}_1 - \hat{p}_2)}{\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}}$ (**B**) $t = \dfrac{(\bar{x}_1 - \bar{x}_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$ (**C**) $t = \dfrac{\bar{x}_d}{\left(\frac{s_d}{\sqrt{n}}\right)}$

Solution: Choice (A) makes no sense since our response variable is quantitative (there are no sample proportions here). Choice (B) is correct for an UNPAIRED analysis. Choice (C) is correct for a PAIRED analysis.

FREE-RESPONSE QUESTION 4 (5 points)

After plugging in his data to the appropriate applet, Anders obtained the following 95% confidence interval for the parameter of interest: **(–5.34, –2.23)**. Interpret this interval, in context.

Solution: We are 95% confident that the population average enjoyment rating among partygoers at the party playing techno music are between 2.23 and 5.34 points LOWER than the population average enjoyment rating among partygoers at the playing pop music.